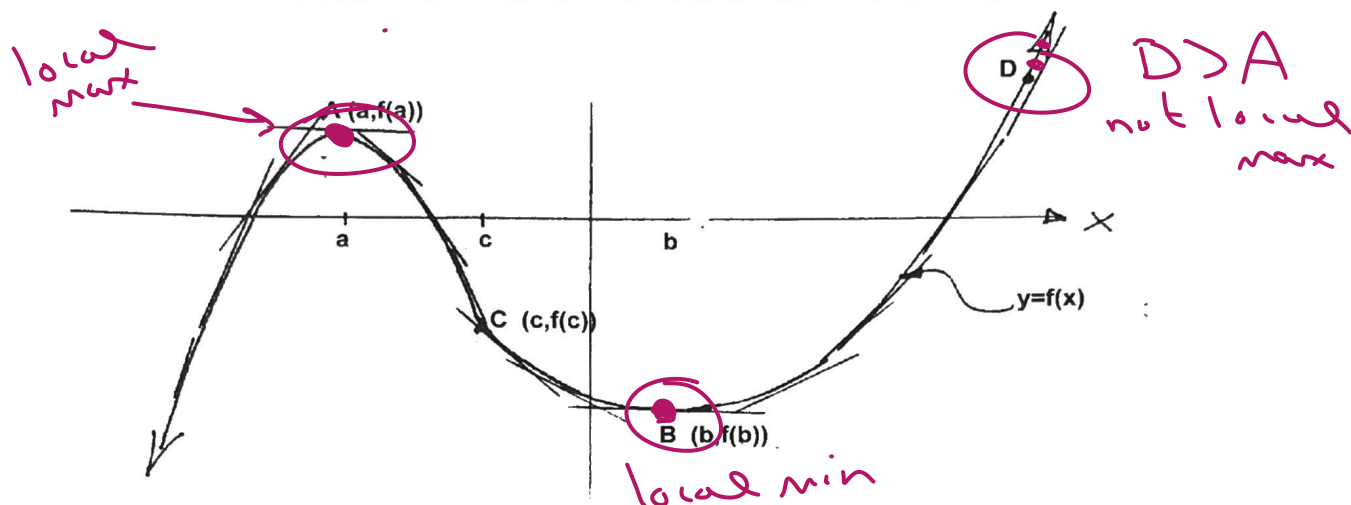


Maximum and Minimum Values

Knowledge of the slope of the tangent line to a curve (function) allows us to pinpoint locations of local maxima or minima. Also known as local extrema.



Point A (on the graph above) is termed a **relative or local maximum** point because in A's vicinity there are no points higher. D is higher but it is not in the vicinity of A. Similarly B is known as a **relative or local minimum** point, because in B's vicinity there is no point lower.

Critical Number:

A number c in the domain of f is a critical number of f if either $f'(c) = 0$ or $f'(c)$ does not exist. The point $(c, f(c))$ is called a **critical point**.

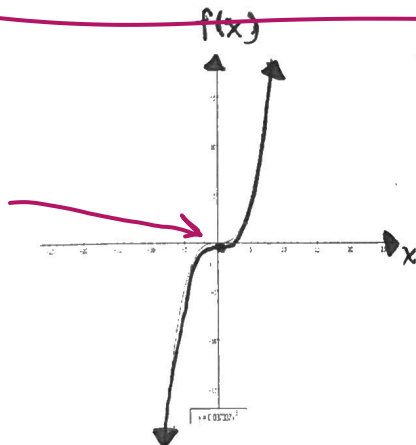
Handwritten notes: *L. Max/min.* (pointing to $f'(c) = 0$), *cusp* (pointing to $f'(c)$ does not exist).

Fermat's Theorem:

If f has a local maximum or minimum value at $x = c$, then either $f'(c) = 0$ or $f'(c)$ does not exist.

1st Derivative Test

For a local maximum to occur, $f'(x)$ must change from positive to negative.
For a local minimum to occur, $f'(x)$ must change from negative to positive.



Note: Note the sign of $f'(x)$ must change for a local extrema.

Example: Consider $f(x) = x^3$. ($f'(x) = 3x^2$)

When $x = 0$, $f'(x) = 0 \leftarrow m_t = 0$

Yet $(0,0)$ is obviously not a relative maximum nor minimum point. It may at best be described as a "flat spot". Mathematically, it is called a "point of inflection" (POI).

$$x^n + y^n = z^n \quad \begin{matrix} n \in \mathbb{Z}^+ \\ n \geq 3 \\ x, y, z \in \mathbb{Z}^+ \end{matrix}$$

Maximum and Minimum Values

Example 1: Determine the coordinates of any local extrema for the function $f(x) = x^4 - x^3$.

$$f(x) = x^3(x-1)$$

0,1

$$f'(x) = 4x^3 - 3x^2$$

$$= x^2(4x-3) \Rightarrow \text{critical \#s @ } x=0, \frac{3}{4}$$

Interval	$(-\infty, 0)$	$(0, \frac{3}{4})$	$(\frac{3}{4}, \infty)$
$\rightarrow x^2$	+	+	+
$\rightarrow 4x-3$	-	-	+
$f'(x)$	-	-	+
$f(x)$	↓	↓	↑

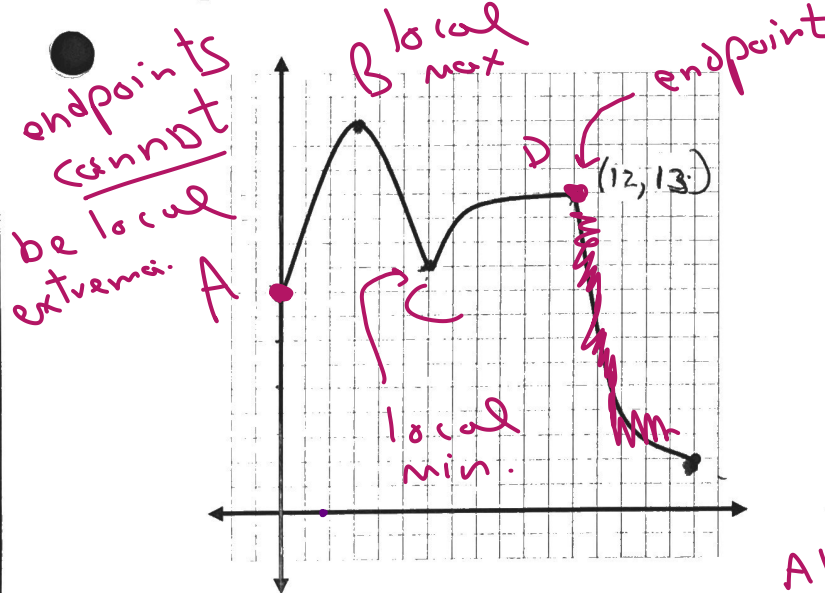
* $x=0$ is not a local extrema

* local min. @ $(\frac{3}{4}, -\frac{27}{256})$

To find absolute maximum or minimum points, you need to also test the endpoints of the domain.



Example 2: Determine the local and absolute extrema for each function:



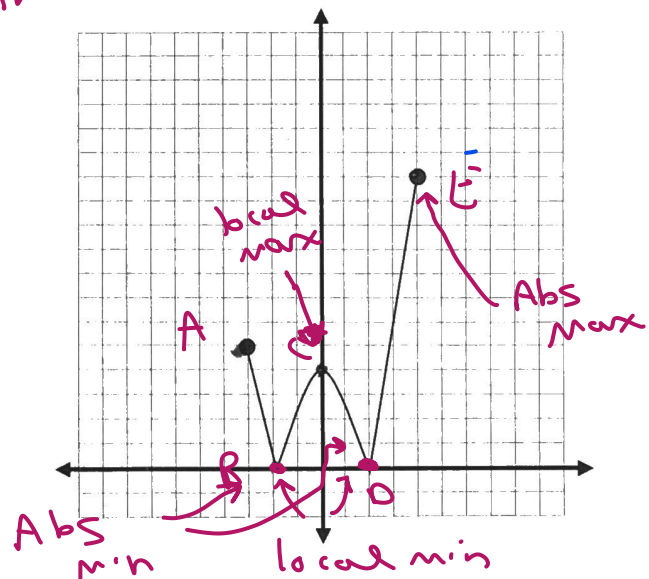
$$y = f(x) \text{ for } x \in [0, 12]$$

Local Max: (3, 16)

Local min: (6, 10)

Abs. max: (3, 16)

Abs. min: (0, 9)



$$y = |x^2 - 4| \text{ for } x \in [-3, 4]$$

Local max: (0, 4)

Local min: (-2, 0) & (2, 0)

Abs. max: (4, 12)

Abs. min: (-2, 0) & (2, 0)

Maximum and Minimum Values

Example 3: Determine the absolute maximum and absolute minimum values of the function

$$f(x) = 2x^4 - 4x^2 + 4 \text{ for } x \in [-2, 0.5].$$

$$f'(x) = 8x^3 - 8x$$

$$= 8x(x^2 - 1)$$

$\Rightarrow$ critical #'s @ $x = 0, \pm 1$.

1 is not a critical #.

not in domain!

Interval	$[-2, -1]$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
x	-	-	+	+
$x^2 - 1$	+	-	-	+
$f'(x)$	-	+	-	+
$f(x)$	↓	↑	↓	↑

not in domain

local min @ $x = -1$

local max @ $x = 0$

Test

min * $f(-1) = 2 \rightarrow$ Absolute min. @ $(-1, 2)$

max * $f(0) = 4 \rightarrow$ Local Max.

* $f(1) = 2 \rightarrow$ NOT IN DOMAIN

* $f(-2) = 20 \rightarrow$ Absolute max. @ $(-2, 20)$

$$* f(0.5) = \frac{25}{8} = 3.125$$

Maximum and Minimum Values

Additional Questions

1. Determine the local max./min. of $f(x) = 3x^5 - 5x^3$.

$$\begin{aligned} f'(x) &= 15x^4 - 15x^2 \\ &= 15x^2(x^2 - 1) \end{aligned}$$

$\therefore$ critical #'s: $x = 0, \pm 1$

Interval	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
x^2	+	+	+	+
$x^2 - 1$	+	-	-	+
$f'(x)$	+	-	-	+
$f(x)$	$\uparrow$	$\downarrow$	$\downarrow$	$\uparrow$
		local max. @ $(-1, 2)$	local min. @ $(1, -2)$	

NOTE:

$(0, 0)$ is not a local max/min.
no change in $f'(x)$

2. Given $f(x) = ax^3 + bx^2 + cx + d$, determine the values of a, b, c and d such that there is a local maximum at $(-2, 13)$ and a local minimum at $(0, 5)$.

$$f'(x) = 3ax^2 + 2bx + c$$

sub. $x = -2$ into $f'(x)$ and set $= 0$.

$$12a - 4b + c = 0 \quad (1)$$

sub. $x = 0$ into $f'(x)$ and set $= 0$

$$\boxed{c = 0}$$

sub. $(0, 5)$ into $f(x)$

$$\boxed{d = 5}$$

sub. $(-2, 13)$ into $f(x)$

$$-8a + 4b + 5 = 13$$

$$-8a + 4b = 8 \quad (2)$$

$$\textcircled{1} + \textcircled{2}$$

$$4a = 8$$

$$\boxed{a = 2}$$

sub. $a = 2$ into $\textcircled{2}$

$$\boxed{b = 6}$$